STAT 581: Large Sample Theory

Handwritten Notes 3, with an Illustrated Guide to Distances

Liang-Cheng Chen

October 19, 21, and 26, 2026

Sources and editorial conventions

This English edited transcription covers all 31 pages of handwritten `../notes/stat581-note3.pdf`, including the October 26 additions on source pp. 21–31. The first 20 pages were transcribed previously. The reference is Yen-Chi Chen, *Lecture 2: Introduction of large sample theory*, Autumn 2026, `Lec2_large.pdf`.

Source-page labels refer to the handwritten PDF. Red boxes identify mathematical corrections; yellow boxes preserve and answer the original five and four newly added explicit TODOs; blue boxes add explanations. Appendix A expands all seven distance families in lecture Appendix B, with original vector diagrams. The diagrams are explanatory examples, not claims about the source drawings.

All formulas use equation or align environments. There is no inline math: TeX treats double-dollar delimiters as display math, not inline math.

October 26 source comparison

Cyan boxes identify omissions relative to the original typed version and give teacher-source locations. Newly recovered classroom examples are ordinary body text with source labels. Existing yellow, red, and blue boxes retain their original meanings. The detailed audit is in `coverage-audit.md`. The later October 26 handwriting update is integrated into the order-calculus and delta-method sections. Source-page references there use the current 31-page student PDF; teacher-source references remain explicitly labeled.

Contents

1 Modes of convergence **4**

1.1 Almost sure convergence and convergence in probability 4

1.2 The typewriter counterexample 5

1.3 Consistency and convergence in distribution 5

2 Distributional distances in the handwritten notes **7**

3 Moment convergence and transformations **8**

3.1 Convergence in a moment norm 8

3.2 Continuous mapping and Slutsky's lemma 9

4 Laws of large numbers and central limit theorems **9**

4.1 The IID results 9

4.2 Triangular arrays: Lindeberg–Feller and Lyapunov 10

5 Berry–Esseen, KS distance, and the bootstrap **11**

5.1 A finite-sample normal approximation 11

5.2 KS versus weak convergence 12

5.3 Empirical bootstrap for the sample mean 12

6 Deterministic and stochastic orders **13**

6.1 Examples and nonuniqueness of rates 14

7 Additional Portmanteau formulations **15**

8 Fixed-design regression as a triangular-array application **15**

9 Stochastic-order calculus and sample variance **16**

9.1 Stochastic algebra and all eight mixed relations 16

9.2 What an order statement does not imply 17

9.3 Density estimation: bias, variability, and bandwidth 18

9.4 Sample variance: expand the remainders, state the assumptions 19

9.5 Gaussian concentration gives more than a stochastic order 20

10 Delta method and asymptotic linearity **21**

10.1 Relative risk: cell proportions, covariance, and inference 22

10.2 Modern approach: expand the ratio directly 24

11 Bootstrap consistency with only a second moment **25**

A Illustrated distributional distances **26**

A.1 Lévy: allow horizontal and vertical CDF error 26

A.2 Prokhorov: enlarge an event and allow missing mass 27

A.3 Bounded Lipschitz: limited range and slope 28

A.4 Kolmogorov–Smirnov: the worst threshold-probability error 29

A.5 Wasserstein: minimum transport cost 30

A.6 Total variation: the best event for distinguishing two laws 31

A.7 Hellinger: square-root density geometry 32

A.8 What converges, and what may still go wrong? 33

A.9 Two examples that separate the distances 34

A.10 A fixed target matters: topology versus uniform equivalence 34

1 Modes of convergence

1.1 Almost sure convergence and convergence in probability

Source: pp. 1–2; lecture Definitions 2.1–2.2 and Theorem 2.4.

Large sample theory studies sequences of random variables. The index may represent sample size, an optimization iteration, or time. For almost sure convergence and convergence in probability, the variables and their limit are defined on a common probability space. For random vectors use the Euclidean norm:

$$X_n \xrightarrow{\text{a.s.}} Z \quad \Longleftrightarrow \quad \mathbb{P}\Big(\lim_{n\to\infty} \|X_n - Z\| = 0\Big) = 1, \tag{1}$$

$$X_n \xrightarrow{P} Z \quad \Longleftrightarrow \quad \forall \varepsilon > 0, \quad \mathbb{P}(\|X_n - Z\| > \varepsilon) \longrightarrow 0. \tag{2}$$

Equivalently, convergence in probability says

$$X_n - Z = o_P(1). \tag{3}$$

This notation alone specifies no numerical rate. A concentration bound is more quantitative; for example,

$$\mathbb{P}(\|X_n - Z\| > \varepsilon) \le b(n, \varepsilon), \qquad b(n, \varepsilon) \longrightarrow 0 \quad \text{for every fixed } \varepsilon > 0. \tag{4}$$

The basic implications and uniqueness statements are

$$X_n \xrightarrow{\text{a.s.}} Z \implies X_n \xrightarrow{P} Z, \tag{5}$$

$$X_n \xrightarrow{\text{a.s.}} Z, \quad X_n \xrightarrow{\text{a.s.}} Y \implies \mathbb{P}(Z = Y) = 1, \tag{6}$$

$$X_n \xrightarrow{P} Z, \quad X_n \xrightarrow{P} Y \implies \mathbb{P}(Z = Y) = 1. \tag{7}$$

Supplement: How Borel–Cantelli enters (source p. 1)

A useful sufficient condition for almost sure convergence is

$$\sum_{n=1}^{\infty} \mathbb{P}(\|X_n - Z\| > \varepsilon) < \infty \quad \text{for every } \varepsilon > 0. \tag{8}$$

Apply the first Borel–Cantelli lemma at reciprocal positive integers and intersect the resulting probability-one events. Independence is unnecessary for this direction. The second lemma requires independence (or an appropriate replacement condition) to infer infinitely many occurrences from a divergent sum; it is not needed for the typewriter example below.

1.2 The typewriter counterexample

Source: p. 2; lecture Example 2.3.

Divide the unit interval into increasingly fine equal pieces, and let each block of the sequence scan all pieces once:

$$U \sim \mathrm{Unif}[0,1], \qquad N_{m,j} = \frac{m(m-1)}{2} + j, \quad 1 \le j \le m, \tag{9}$$

$$X_{N_{m,j}} = \mathbf{1}\left\{ \frac{j-1}{m} \le U < \frac{j}{m} \right\}. \tag{10}$$

For any threshold strictly between zero and one, the exceedance probability is the reciprocal block size:

$$\mathbb{P}(X_{N_{m,j}} > \varepsilon) = \frac{1}{m} \longrightarrow 0, \qquad 0 < \varepsilon < 1. \tag{11}$$

Thus the sequence converges to zero in probability. For every realization strictly below one, exactly one entry in each block equals one. There are also infinitely many zeros, so almost sure convergence to zero fails. The endpoint one is a null event.

Correction: Indexing the blocks

The handwritten second block reuses earlier indices. The triangular indexing above removes that duplication while preserving the construction. The probability of error vanishes, but a fixed sample path keeps encountering errors. These are different statements.

1.3 Consistency and convergence in distribution

Source: pp. 3–4; lecture Definitions 2.5 and Theorems 2.6–2.7.

An estimator is consistent when its estimation error tends to zero in probability:

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \Longleftrightarrow \quad \|\hat{\theta}_n - \theta\| \xrightarrow{P} 0. \tag{12}$$

This does not by itself imply convergence of mean squared error. For example,

$$\mathbb{P}(X_n = n^2) = \frac{1}{n}, \quad \mathbb{P}(X_n = 0) = 1 - \frac{1}{n} \quad \Longrightarrow \quad X_n \xrightarrow{P} 0, \qquad \mathbb{E}X_n^2 = n^3. \tag{13}$$

Convergence in distribution is convergence of laws, rather than closeness of sample paths. In one dimension its CDF characterization is

$$X_n \xrightarrow{d} Z \quad \Longleftrightarrow \quad F_{X_n}(t) \longrightarrow F_Z(t) \quad \text{at every continuity point of } F_Z. \tag{14}$$

Correction: The continuity-point qualification is essential

Source p. 3 omits this qualification. At a jump of the limiting CDF, weak convergence need not imply pointwise CDF convergence. The normal-to-point-mass example in Section 5.2 makes this explicit.

The handwritten Portmanteau formulations are the following equivalent conditions, in a fixed finite-dimensional Euclidean space:

$$X_n \xrightarrow{d} Z; \tag{15}$$

$$\mathbb{E}f(X_n) \longrightarrow \mathbb{E}f(Z) \quad \text{for every bounded continuous } f; \tag{16}$$

$$\mathbb{E}f(X_n) \longrightarrow \mathbb{E}f(Z) \quad \text{for every bounded Lipschitz } f; \tag{17}$$

$$\mathbb{E}e^{it^{\mathsf{T}}X_n} \longrightarrow \mathbb{E}e^{it^{\mathsf{T}}Z} \quad \text{for every } t \in \mathbb{R}^d; \tag{18}$$

$$t^{\mathsf{T}}X_n \xrightarrow{d} t^{\mathsf{T}}Z \quad \text{for every } t \in \mathbb{R}^d. \tag{19}$$

The fourth line is Lévy's continuity theorem; characteristic functions are Fourier transforms of probability measures, which need not have densities. The fifth is the Cramér–Wold device. Also,

$$X_n \xrightarrow{P} Z \implies X_n \xrightarrow{d} Z, \tag{20}$$

$$X_n \xrightarrow{d} Z, \quad X_n \xrightarrow{d} Y \implies \mathrm{Law}(Z) = \mathrm{Law}(Y). \tag{21}$$

Uniqueness of the limiting law does not assert equality of the random variables on a common sample space. Convergence in distribution to a constant, however, does imply convergence in probability to that constant.

TODO resolved: N3-T1: A confidence interval and a test (source p. 3)

Suppose independent identically distributed observations have finite mean and positive finite variance. The CLT gives

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1). \tag{22}$$

Let the standard-normal quantile be defined by

$$\Phi(z_{1-\alpha/2}) = 1 - \alpha/2. \tag{23}$$

Continuity of the normal CDF implies

$$\mathbb{P}\Big(\mu \in [\bar{X}_n - z_{1-\alpha/2}\sigma/\sqrt{n}, \bar{X}_n + z_{1-\alpha/2}\sigma/\sqrt{n}]\Big) \longrightarrow 1 - \alpha. \tag{24}$$

Under the null hypothesis that the mean equals a specified value, replacing the mean in the pivot by that value gives a two-sided test: reject when the absolute pivot exceeds this quantile. Its type-I error tends to the nominal level. A consistent standard deviation estimate may replace the known standard deviation by Slutsky's lemma.

2 Distributional distances in the handwritten notes

Source: pp. 5–7; lecture Appendix B, Definitions 2.36–2.44.

Use probability measures for events and CDFs for one-dimensional threshold probabilities. The three metrics that metrize weak convergence are

$$d_L(F,G) = \inf\{\varepsilon > 0 : F(t-\varepsilon) - \varepsilon \le G(t) \le F(t+\varepsilon) + \varepsilon \text{ for all } t\}, \tag{25}$$

$$d_{\mathrm{Pr}}(P,Q) = \inf\{\varepsilon > 0 : P(A) \le Q(A^\varepsilon) + \varepsilon \text{ for all Borel } A\}, \tag{26}$$

$$A^\varepsilon = \{x : \inf_{y \in A} \|x - y\| < \varepsilon\}, \tag{27}$$

$$d_{\mathrm{BL}}(P,Q) = \sup_{\|f\|_\infty \le 1,\, \mathrm{Lip}(f) \le 1} |\mathbb{E}_P f - \mathbb{E}_Q f|, \tag{28}$$

$$\mathrm{Lip}(f) = \sup_{x \ne y} \frac{|f(x) - f(y)|}{\|x - y\|}, \qquad \|f\|_\infty = \sup_x |f(x)|. \tag{29}$$

Here the Prokhorov definition is the lecture's one-sided formulation; for probability measures it is equivalent to imposing both directions. Appendix A.2 uses the symmetric formulation for its geometric explanation. Lévy distance applies to real-valued distributions; the other two apply on Euclidean spaces. For a fixed real-valued limiting law,

$$X_n \xrightarrow{d} Z \iff d_L(F_{X_n}, F_Z) \to 0 \iff d_{\mathrm{Pr}}(P_{X_n}, P_Z) \to 0 \iff d_{\mathrm{BL}}(P_{X_n}, P_Z) \to 0. \tag{30}$$

Correction: A supremum, not an infimum (source pp. 5 and 7)

The handwritten BL and Wasserstein dual formulas use an infimum over test functions. That would always give zero, because the zero function is admissible. The correct operation is the supremum: choose the test function that best separates the two distributions. Wasserstein's transport formulation instead uses an infimum over couplings.

Weak convergence does not control unbounded observables. The handwritten contamination example can be written as

$$P_n = \left(1 - \frac{1}{n}\right) N(0,1) + \frac{1}{n}\delta_{n^2}, \qquad P_n \Rightarrow N(0,1), \qquad \mathbb{E}_{P_n} X = n. \tag{31}$$

Couple the mixture to a shared standard normal variable, replacing it only on the contamination event. The discrepancy probability is at most the mixture weight, so convergence in probability also holds, while the mean diverges. Every bounded test function sees a discrepancy at most twice its bound times that weight. Threshold indicators are bounded but not Lipschitz; weak convergence controls them only at limiting CDF continuity points.

For laws with finite first absolute moments, the Wasserstein dual is

$$W_1(P,Q) = \sup_{\mathrm{Lip}(f) \le 1} |\mathbb{E}_P f - \mathbb{E}_Q f|. \tag{32}$$

It removes the BL amplitude bound. The precise moment equivalence is

$$W_1(P_n, P) \to 0 \quad \iff \quad P_n \Rightarrow P \text{ and } \int \|x\|\, dP_n \to \int \|x\|\, dP. \tag{33}$$

Correction: Signed means are insufficient (source p. 7)

Replace convergence of signed means by convergence of first absolute moments. For example, the following laws have zero means and converge weakly to a point mass, yet their Wasserstein distance does not vanish:

$$P_n = (1 - 1/n)\delta_0 + (1/2n)(\delta_n + \delta_{-n}), \qquad \mathbb{E}_{P_n} X = 0, \quad W_1(P_n, \delta_0) = 1. \tag{34}$$

TODO resolved: N3-T2: BL intuition (source p. 5)

A BL function is a measuring instrument with a bounded reading and bounded slope. It cannot assign an arbitrarily large score to a tiny amount of faraway mass, or change its reading abruptly across two nearby points. Its worst possible expectation discrepancy measures weak convergence. See the illustrated BL page in Appendix A.3.

TODO resolved: N3-T3: Why Wasserstein controls expectation (source p. 7)

A linear coordinate function is Lipschitz but unbounded, so it is allowed in the Wasserstein dual and excluded from the BL class. In one dimension,

$$|\mathbb{E}_P X - \mathbb{E}_Q X| \le W_1(P, Q). \tag{35}$$

The absolute-value function is also 1-Lipschitz, giving

$$|\mathbb{E}_P |X| - \mathbb{E}_Q |X|| \le W_1(P, Q). \tag{36}$$

Thus vanishing transport cost forces both discrepancies to vanish. The converse needs weak convergence and absolute moments as in (33); convergence of signed means alone is too weak.

3 Moment convergence and transformations

3.1 Convergence in a moment norm

Source: p. 8; lecture Definition 2.9 and Theorem 2.10.

For a positive exponent, define moment convergence by

$$X_n \xrightarrow{L^p} Z \quad \iff \quad \mathbb{E}\|X_n - Z\|^p \to 0. \tag{37}$$

On a probability space, a larger moment controls a smaller one:

$$s > p > 0 \quad \Longrightarrow \quad \left(\mathbb{E}\|X_n - Z\|^p\right)^{1/p} \le \left(\mathbb{E}\|X_n - Z\|^s\right)^{1/s}. \tag{38}$$

Moreover, Markov's inequality gives

$$\mathbb{P}(\|X_n - Z\| > \varepsilon) \leq \frac{\mathbb{E}\|X_n - Z\|^p}{\varepsilon^p} \to 0. \tag{39}$$

This implication holds for every positive exponent, including those below one (where the usual moment expression is not a norm). A limit in this sense is unique almost surely. Squared moment convergence to a parameter is exactly vanishing mean squared error.

3.2 Continuous mapping and Slutsky's lemma

Source: pp. 8–9; lecture Theorems 2.12–2.13.

Suppose a measurable map is continuous at every point of a set receiving probability one under the limiting variable. Applying it preserves each of almost sure convergence, convergence in probability, and convergence in distribution, respectively:

$$X_n \xrightarrow{\star} Z \quad \Longrightarrow \quad f(X_n) \xrightarrow{\star} f(Z), \qquad \star \in \{\text{a.s.}, P, d\}. \tag{40}$$

Slutsky's lemma combines a distributional limit with a constant probability limit:

$$X_n \xrightarrow{d} Z, \quad Y_n \xrightarrow{P} c \quad \Longrightarrow \quad X_n + Y_n \xrightarrow{d} Z + c, \tag{41}$$

$$Y_n X_n \xrightarrow{d} cZ \quad \text{when } Y_n \text{ is scalar}, \tag{42}$$

$$X_n / Y_n \xrightarrow{d} Z/c \quad \text{when } Y_n \text{ is scalar and } c \neq 0. \tag{43}$$

Addition requires matching dimensions. No independence assumption is needed. The ratio assertion is central to studentization: a consistent positive standard deviation estimate can replace the unknown population standard deviation in a CLT pivot.

4 Laws of large numbers and central limit theorems

4.1 The IID results

Source: p. 10; lecture Theorems 2.14–2.15.

For independent identically distributed observations, finite first absolute moment gives the strong law, hence also the weak law:

$$\mathbb{E}|X_1| < \infty \quad \Longrightarrow \quad \bar{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}X_1, \qquad \bar{X}_n \xrightarrow{P} \mathbb{E}X_1. \tag{44}$$

For random vectors with finite second moment,

$$\mu = \mathbb{E}X_1, \qquad \Sigma = \mathbb{E}[(X_1 - \mu)(X_1 - \mu)^{\mathsf{T}}], \tag{45}$$

$$\mathbb{E}\|X_1\|^2 < \infty \quad \Longrightarrow \quad \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N_d(0, \Sigma). \tag{46}$$

A singular covariance is allowed; the limiting normal may be degenerate. The right side of a convergence statement must be a fixed law, not a normal law whose covariance still depends on the sample size.

4.2 Triangular arrays: Lindeberg–Feller and Lyapunov

Source: pp. 11–12 (October 21); lecture Theorem 2.17.

For each row, let the observations be independent, without requiring identical distributions or independence between rows:

$$X_{n,1}, \dots, X_{n,k_n}, \qquad k_n \longrightarrow \infty, \tag{47}$$

$$\mu_{n,j} = \mathbb{E}X_{n,j}, \quad \sigma_{n,j}^2 = \operatorname{Var}(X_{n,j}) < \infty, \tag{48}$$

$$s_n^2 = \sum_{j=1}^{k_n} \sigma_{n,j}^2 > 0, \qquad Y_{n,j} = \frac{X_{n,j} - \mu_{n,j}}{s_n}. \tag{49}$$

The denominator is the row's total standard deviation, not its total variance. Either of the following sufficient conditions implies a normal limit:

$$\text{Lindeberg:} \quad \sum_{j=1}^{k_n} \mathbb{E}[Y_{n,j}^2 \mathbf{1}\{|Y_{n,j}| > \varepsilon\}] \longrightarrow 0 \quad \text{for every } \varepsilon > 0; \tag{50}$$

$$\text{Lyapunov:} \quad \sum_{j=1}^{k_n} \mathbb{E}|Y_{n,j}|^{2+\delta} \longrightarrow 0 \quad \text{for some } \delta > 0. \tag{51}$$

The conclusion is

$$\sum_{j=1}^{k_n} Y_{n,j} \xrightarrow{d} N(0,1). \tag{52}$$

Lyapunov implies Lindeberg because each truncated second moment is bounded by the higher moment divided by the threshold to the extra power. For an IID row with common positive variance and row length equal to the sample size, the normalization becomes

$$s_n = \sqrt{n}\sigma, \qquad \sum_{j=1}^{n} Y_{n,j} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}. \tag{53}$$

This framework also proves normality of regression estimators by expressing them as weighted sums of independent errors. A design condition must ensure no individual weighted observation dominates; the handwritten reference to OLS is a motivation, not a proof for every design.

5 Berry–Esseen, KS distance, and the bootstrap

5.1 A finite-sample normal approximation

Source: pp. 13–14; lecture Appendix A, Theorem 2.34 (Berry–Esseen).

For IID real observations, assume positive finite variance and a finite third absolute centered moment. For some universal finite constant,

$$W_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}, \qquad \rho_3 = \mathbb{E}|X_1 - \mu|^3 < \infty, \tag{54}$$

$$\sup_t |\mathbb{P}(W_n \le t) - \Phi(t)| \le e_n := \frac{C\rho_3}{\sigma^3\sqrt{n}}. \tag{55}$$

The bound holds at every sample size. It is informative when the bound is small; a bound exceeding one is valid but uninformative.

Correction: Center the third absolute moment (source p. 13)

The source uses the uncentered third absolute moment without assuming a zero mean. The standard Berry–Esseen expression uses the centered moment. Finite uncentered third absolute moment is sufficient for its finiteness, but the displayed numerical bound must use the correct quantity.

TODO resolved: N3-T4: Explain coverage and type-I error (source p. 14)

For a one-sided normal critical value, the KS bound yields

$$|\mathbb{P}(W_n \le z_{1-\alpha}) - (1-\alpha)| \le e_n. \tag{56}$$

Consequently, the upper confidence interval for the mean has coverage within this error of the target:

$$\mathbb{P}\big(\mu \le \bar{X}_n + z_{1-\alpha}\sigma/\sqrt{n}\big) \in [1-\alpha-e_n, 1-\alpha+e_n]. \tag{57}$$

For the displayed upper interval apply the same KS bound at the negative critical value; normal symmetry gives the stated target. Intersect bounds with the unit interval when necessary. Similarly, rejecting the one-sided null when the pivot exceeds the critical value gives type-I error within the same error of the nominal level. A two-sided event uses two CDF endpoints, so its coverage error is at most twice this bound.

These finite-sample bounds concern the known-variance pivot. Substituting an estimated standard deviation preserves asymptotic validity under consistency, but the same finite-sample bound does not follow from Slutsky alone.

5.2 KS versus weak convergence

Source: pp. 14–15; lecture Definitions 2.40 and Theorem 2.42.

$$d_{\mathrm{KS}}(F, G) = \sup_t |F(t) - G(t)|. \tag{58}$$

KS convergence implies weak convergence. The converse holds if the fixed limiting CDF is continuous everywhere (Pólya's theorem).

TODO resolved: N3-T5: Weak convergence without KS convergence (source p. 15)

Let the normal variance decrease to zero:

$$X_n \sim N(0, 1/n), \qquad X_n \xrightarrow{d} 0, \qquad F_0(t) = \mathbf{1}\{t \geq 0\}. \tag{59}$$

At zero the normal CDF is one half and the limiting CDF is one. Indeed,

$$d_{\mathrm{KS}}(F_{X_n}, F_0) = \frac{1}{2} \quad \text{for every } n. \tag{60}$$

The limiting CDF is discontinuous at zero, so Pólya's continuity hypothesis fails. Weak convergence correctly ignores this discontinuity point.

5.3 Empirical bootstrap for the sample mean

Source: pp. 16–17; lecture Appendix A.

Given IID data with finite mean and positive variance, form the empirical law and sample conditionally independently with replacement from it:

$$\widehat{P}_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}, \qquad \widehat{F}_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \leq t\}, \tag{61}$$

$$X_1^*, \ldots, X_n^* \mid X_1, \ldots, X_n \overset{\text{iid}}{\sim} \widehat{P}_n, \tag{62}$$

$$\bar{X}_n^* = \frac{1}{n}\sum_{i=1}^{n} X_i^*, \qquad \hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n} (X_i - \bar{X}_n)^2. \tag{63}$$

Bootstrap fluctuations about the observed mean approximate sampling fluctuations about the population mean. Under a finite third absolute moment, the rate statement in the handwriting has the rigorous form

$$\sup_t \left| \mathbb{P}^*\left(\frac{\sqrt{n}(\bar{X}_n^* - \bar{X}_n)}{\hat{\sigma}_n} \leq t \right) - \mathbb{P}\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq t \right) \right| = O_P(n^{-1/2}). \tag{64}$$

Here the starred probability is conditional on the observed data. On the rare event of zero empirical variance, define the starred pivot arbitrarily; its probability tends to zero under the positive population variance assumption.

To see the rate, apply Berry–Esseen conditionally to the empirical law:

$$\sup_t |\mathbb{P}^*(W_n^* \le t) - \Phi(t)| \le \frac{C\hat{\rho}_{3,n}}{\hat{\sigma}_n^3\sqrt{n}}, \qquad \hat{\rho}_{3,n} = \frac{1}{n}\sum_{i=1}^n |X_i - \bar{X}_n|^3. \tag{65}$$

The empirical third centered moment is bounded in probability and the empirical standard deviation converges to a positive constant. Apply the ordinary Berry–Esseen bound to the unstarred pivot and use the triangle inequality to obtain (64).

Correction: What this bootstrap theorem says

The source does not state its moment and nondegeneracy assumptions. They are needed for the displayed rate. The second probability is a sampling CDF, not itself "the true CI coverage." Coverage follows only after choosing an interval and relating its endpoints to the approximated CDF. The bootstrap pivot above divides by the original empirical standard deviation, not a newly computed standard deviation in every bootstrap sample.

6 Deterministic and stochastic orders

Source: pp. 17–20; lecture Section 2.5.

For deterministic sequences with nonzero scaling terms eventually,

$$a_n = o(1) \iff a_n \to 0, \tag{66}$$
$$a_n = o(b_n) \iff a_n/b_n \to 0, \tag{67}$$
$$a_n = O(1) \iff \exists C < \infty,\ N : \ |a_n| \le C \text{ for all } n \ge N, \tag{68}$$
$$a_n = O(b_n) \iff a_n/b_n = O(1). \tag{69}$$

The constant must work for every sufficiently large index; choosing a different unrestricted constant at every index would be vacuous.

For random sequences and positive deterministic scales, define

$$X_n = o_P(1) \iff X_n \xrightarrow{P} 0, \tag{70}$$
$$X_n = o_P(a_n) \iff X_n/a_n \xrightarrow{P} 0, \tag{71}$$
$$X_n = O_P(1) \iff \forall \varepsilon > 0\ \exists M < \infty,\ N : \sup_{n \ge N} \mathbb{P}(\|X_n\| > M) < \varepsilon, \tag{72}$$
$$X_n = O_P(a_n) \iff X_n/a_n = O_P(1). \tag{73}$$

Boundedness in probability concerns tails uniformly for sufficiently large indices. It does not require a fixed almost sure bound. In finite-dimensional Euclidean space it is equivalent to tightness of the sequence of laws; the finitely many initial distributions do not affect this property.

A fixed Cauchy law also gives a tight sequence, despite having no finite mean or variance. This classroom example (teacher handnotes October 21, p. 9) emphasizes that stochastic boundedness is a tail-probability property, not a moment assumption:

$$X_n \sim \text{Cauchy}(1,1) \quad \text{for every } n \quad \Longrightarrow \quad X_n = O_P(1). \tag{74}$$

6.1 Examples and nonuniqueness of rates

Source: pp. 19–20.

$$X_n \sim N(0, 1/n) \quad \Longrightarrow \quad X_n = O_P(n^{-1/2}), \tag{75}$$

$$\mathbb{E}X_n = 0, \quad \text{Var}(X_n) = a_n^2 \quad \Longrightarrow \quad \mathbb{P}(|X_n| > Ma_n) \le M^{-2}, \quad X_n = O_P(a_n). \tag{76}$$

A rate statement is an upper bound, not necessarily the sharpest rate. For example,

$$X_n = o_P(1) \implies X_n = O_P(1), \tag{77}$$

$$X_n = o_P(1) \implies X_n = o_P(n) \implies X_n = O_P(n^2), \tag{78}$$

$$X_n = O_P(1) \implies X_n = O_P(n) \implies X_n = O_P(n^2). \tag{79}$$

Convergence in law implies boundedness in probability, and boundedness in probability guarantees a weakly convergent subsequence:

$$X_n \xrightarrow{d} X \quad \Longrightarrow \quad X_n = O_P(1), \tag{80}$$

$$X_n = O_P(1) \quad \Longrightarrow \quad \exists n_k \uparrow \infty, \ \exists P : \ \text{Law}(X_{n_k}) \Rightarrow P. \tag{81}$$

Correction: A false converse (source p. 19)

The handwritten equivalence between full-sequence weak convergence and boundedness in probability must be a one-way implication. A deterministic sequence alternating between zero and one is bounded, but its laws do not converge. Its even and odd subsequences have different limits.

For IID observations with zero mean and positive finite variance, LLN and CLT yield the final handwritten example:

$$\bar{X}_n = o_P(1), \qquad \frac{\sqrt{n}\bar{X}_n}{\sigma} \xrightarrow{d} N(0,1), \qquad \bar{X}_n = O_P(n^{-1/2}). \tag{82}$$

For a nonzero mean these statements apply to the centered sample mean.

7 Additional Portmanteau formulations

Missing from original typed notes: G2: the omitted set and semicontinuity criteria

Lecture 2, Theorem 2.6, p. 2-3; teacher handnotes October 19, p. 6, explicitly refers to additional lecture formulations. Original Notes 3 lists only the continuous-function, characteristic-function, and projection versions. The lecture's item 7 has a typo: the set must be open.

Weak convergence is equivalent to each of the following criteria:

$$\limsup_n \mathbb{P}(X_n \in C) \le \mathbb{P}(Z \in C) \qquad \text{for every closed set } C, \tag{83}$$

$$\liminf_n \mathbb{P}(X_n \in G) \ge \mathbb{P}(Z \in G) \qquad \text{for every open set } G, \tag{84}$$

$$\mathbb{P}(X_n \in A) \to \mathbb{P}(Z \in A) \qquad \text{if } \mathbb{P}(Z \in \partial A) = 0. \tag{85}$$

Equivalently, the limsup expectation inequality holds for every bounded upper semicontinuous function, and the liminf inequality holds for every bounded lower semicontinuous function. One-sided bounded versions also hold with extended expectations interpreted appropriately.

For a concrete failure of probability convergence, take independent standard normal variables on one common space:

$$X_n \overset{\text{iid}}{\sim} N(0,1), \quad Z \sim N(0,1) \text{ independent of all } X_n. \tag{86}$$

All laws already equal the target law, but the difference has the same nondegenerate normal law at every index, so it cannot converge to zero in probability. This supplies the mechanism in lecture Example 2.8.

8 Fixed-design regression as a triangular-array application

Missing from original typed notes: G3: the OLS proof was only mentioned

Lecture 2, Example 2.18, pp. 2-6–2-8; teacher handnotes October 21, p. 3. The original Notes 3 gives the motivation but omits the design condition and the proof. The following corrects transpose and normalization typos in the lecture's derivation.

For a fixed-dimensional full-rank design with row vectors, assume independent identically distributed errors of mean zero and positive finite variance:

$$Y_n = X_n\beta + e_n, \qquad \operatorname{Var}(e_i) = \sigma^2, \tag{87}$$

$$\hat{\beta} - \beta = (X_n^{\mathsf{T}} X_n)^{-1} X_n^{\mathsf{T}} e_n, \tag{88}$$

$$a_{n,i} = (X_n^\top X_n)^{-1/2} x_{n,i}, \qquad \sum_i a_{n,i} a_{n,i}^\top = I_p. \tag{89}$$

A sufficient no-dominant-observation condition is vanishing maximum leverage:

$$\max_i h_{n,ii} = \max_i \|a_{n,i}\|^2 \to 0. \tag{90}$$

For a fixed nonzero vector, the normalized scalar summands have total variance one. Their Lindeberg sum is bounded by a common tail second moment:

$$b_{n,i} = \frac{t^\top a_{n,i}}{\|t\|}, \quad \sum_i b_{n,i}^2 = 1, \quad m_n = \max_i |b_{n,i}| \to 0, \tag{91}$$

$$\sum_i \mathbb{E}\left[\frac{b_{n,i}^2 e_i^2}{\sigma^2} \mathbf{1}\{|b_{n,i} e_i| > \varepsilon\sigma\}\right] \le \frac{1}{\sigma^2} \mathbb{E}[e_1^2 \mathbf{1}\{|e_1| > \varepsilon\sigma / m_n\}] \to 0. \tag{92}$$

Cramér–Wold therefore gives

$$(X_n^\top X_n)^{1/2} (\hat{\beta} - \beta) \xrightarrow{d} N_p(0, \sigma^2 I_p). \tag{93}$$

If the normalized Gram matrix tends to a positive definite matrix, then

$$n^{-1} X_n^\top X_n \to Q > 0 \quad \Longrightarrow \quad \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N_p(0, \sigma^2 Q^{-1}). \tag{94}$$

Symmetry of the error law is unnecessary. Finite variance and the design condition supply the needed Lindeberg control.

9 Stochastic-order calculus and sample variance

Missing from original typed notes: G4: order algebra and a worked variance estimator

Lecture 2, Proposition 2.25 and Examples 2.26–2.28, pp. 2-10–2-11, were absent from the original typed notes and supplied teacher handnotes. The updated student notes now cover these topics on pp. 21–27.

Student source: pp. 21–27, October 26; lecture Proposition 2.25 and Examples 2.26–2.28.

9.1 Stochastic algebra and all eight mixed relations

The six handwritten rules on p. 21 are, with no independence required,

$$o_P(1) + o_P(1) = o_P(1), \qquad O_P(1) + O_P(1) = O_P(1), \tag{95}$$

$$O_P(1) + o_P(1) = O_P(1), \qquad O_P(1) o_P(1) = o_P(1), \tag{96}$$

$$[1 + o_P(1)]^{-1} = 1 + o_P(1), \qquad o_P(1) \subseteq O_P(1). \tag{97}$$

The inverse therefore is bounded in probability, but does not converge to zero. Define it arbitrarily on any vanishing-probability event where its denominator is zero. The notation describes classes of sequences: repeated order symbols need not denote the same random remainder.

TODO resolved: Complete the eight relations (source p. 22)

For deterministic small/bounded sequences and random small/tight sequences, Lecture 2 Proposition 2.25 gives all four sums and all four products:

$$o_P(1) + o(1) = o_P(1), \qquad o_P(1) + O(1) = O_P(1), \tag{98}$$
$$O_P(1) + o(1) = O_P(1), \qquad O_P(1) + O(1) = O_P(1), \tag{99}$$
$$o_P(1)o(1) = o_P(1), \qquad o_P(1)O(1) = o_P(1), \tag{100}$$
$$O_P(1)o(1) = o_P(1), \qquad O_P(1)O(1) = O_P(1). \tag{101}$$

The blank lines in the handwriting do not specify an ordering; the completed list follows the lecture. The mnemonic "drop P, do algebra, restore P" helps recall these rules, but the definitions justify them.

For example, a product of a tight factor and a factor converging to zero converges to zero because, for any positive threshold and cutoff,

$$\mathbb{P}(|U_n V_n| > \varepsilon) \le \mathbb{P}(|U_n| > M) + \mathbb{P}(|V_n| > \varepsilon/M). \tag{102}$$

First choose the cutoff to control the tight factor, then take the sample size large. Independence is unnecessary. For positive deterministic scales,

$$O_P(a_n) + O_P(b_n) = O_P(a_n + b_n), \tag{103}$$
$$O_P(a_n)O_P(b_n) = O_P(a_n b_n), \tag{104}$$
$$o_P(a_n)O_P(b_n) = o_P(a_n b_n), \tag{105}$$
$$[c + o_P(1)]^{-1} = c^{-1} + o_P(1), \qquad c \neq 0. \tag{106}$$

9.2 What an order statement does not imply

Student source: pp. 23 and 27. Deterministic bounds and bounds in probability differ. The exact classroom example is

$$X_n \sim N(0, n^{-2}), \quad a_n = 1, \qquad X_n + a_n \sim N(1, n^{-2}), \qquad X_n + a_n = O_P(1). \tag{107}$$

Every finite-sample law has unbounded support, so there is no deterministic uniform bound over all possible outcomes. It nevertheless converges to one in probability. In fact a common coupling can even be almost surely eventually bounded; that is distinct from a deterministic bound.

The second handwritten example is

$$X_n \sim \begin{cases} N(0,1), & n \text{ even}, \\ \mathrm{Unif}[-1,1], & n \text{ odd}. \end{cases} \tag{108}$$

This sequence is tight, but its two subsequences have different laws, so there is no full-sequence weak limit. Nor does tightness control moments: a fixed Cauchy law has no finite mean, while the rare-spike example gives

$$\mathbb{P}(X_n = n^2) = 1/n, \quad \mathbb{P}(X_n = 0) = 1 - 1/n, \qquad X_n \xrightarrow{P} 0, \quad \mathbb{E}X_n = n \to \infty. \tag{109}$$

These are different failures: undefined moments versus diverging finite moments. An order bound alone rules out neither.

9.3 Density estimation: bias, variability, and bandwidth

TODO resolved: Supply the KDE versus histogram example (source pp. 23–24)

The handwriting requests a bias–variability decomposition and sketches small Gaussian kernels placed at the observations. No worked teacher example is present in the supplied teacher PDFs. The following is an editorial completion, with the assumptions stated explicitly.

Let the observations be IID with density, and consider a fixed evaluation point and a deterministic bandwidth:

$$\hat{p}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right), \tag{110}$$

$$\hat{p}_h(x) - p(x) = \{\mathbb{E}\hat{p}_h(x) - p(x)\} + \{\hat{p}_h(x) - \mathbb{E}\hat{p}_h(x)\}. \tag{111}$$

Assume the density is bounded and globally Lipschitz, the kernel is nonnegative and integrates to one, and the following integrals are finite:

$$\int |u| K(u)\, du < \infty, \qquad R(K) = \int K(u)^2\, du < \infty, \tag{112}$$

$$|\mathbb{E}\hat{p}_h(x) - p(x)| \le Lh \int |u| K(u)\, du = O(h), \tag{113}$$

$$\mathrm{Var}(\hat{p}_h(x)) \le \frac{\|p\|_\infty R(K)}{nh}. \tag{114}$$

Chebyshev's inequality yields precisely the requested pointwise rate:

$$\hat{p}_h(x) - p(x) = O(h) + O_P((nh)^{-1/2}), \qquad h \to 0, \quad nh \to \infty. \tag{115}$$

For a fixed histogram bin of width equal to the bandwidth and containing the evaluation point, the same Lipschitz assumption gives

$$\hat{p}_{\text{hist}}(x) = \frac{1}{nh} \sum_i \mathbf{1}\{X_i \in B_h(x)\}, \tag{116}$$

$$|\mathbb{E}\hat{p}_{\text{hist}}(x) - p(x)| \le Lh, \qquad \operatorname{Var}(\hat{p}_{\text{hist}}(x)) \le \frac{\|p\|_\infty}{nh}. \tag{117}$$

Thus the two estimators share this conservative rate. Balancing the two terms suggests a bandwidth of order the inverse cube root of sample size and a pointwise error of that same order. The histogram is piecewise constant; Gaussian KDE is smooth.

Supplement: When a symmetric kernel improves the bias

If the density has a bounded continuous second derivative, a symmetric kernel with finite second moment cancels the first-order term:

$$\mu_2(K) = \int u^2 K(u)\, du, \qquad \int uK(u)\, du = 0, \tag{118}$$

$$\mathbb{E}\hat{p}_h(x) - p(x) = \tfrac{1}{2} h^2 \mu_2(K) p''(x) + o(h^2), \tag{119}$$

$$\hat{p}_h(x) - p(x) = O(h^2) + O_P((nh)^{-1/2}). \tag{120}$$

Balancing now suggests a bandwidth of order the inverse fifth root of sample size and pointwise error of order the inverse two-fifths power. These are pointwise statements; boundaries and uniform errors require additional analysis. The original first-order bias is valid under its weaker smoothness assumption, rather than a universal sharp KDE rate.

9.4 Sample variance: expand the remainders, state the assumptions

Student source: pp. 25–27. The red handwritten reminder says to work backward from the desired result, identify each theorem, and check its assumptions before applying it. Here, IID observations with finite fourth moment allow a CLT for the empirical second moment as well as the mean. Write

$$\mu = \mathbb{E}X_1, \quad m_2 = \mathbb{E}X_1^2, \quad \sigma^2 = m_2 - \mu^2, \tag{121}$$

$$M_n = n^{-1} \sum_i X_i^2 = m_2 + R_{2n}, \qquad R_{2n} = O_P(n^{-1/2}), \tag{122}$$

$$\bar{X}_n = \mu + R_{1n}, \qquad R_{1n} = O_P(n^{-1/2}), \tag{123}$$

$$\bar{X}_n^2 = \mu^2 + 2\mu R_{1n} + R_{1n}^2, \qquad R_{1n}^2 = O_P(n^{-1}), \tag{124}$$

$$M_n - \bar{X}_n^2 = \sigma^2 + R_{2n} - 2\mu R_{1n} - R_{1n}^2 = \sigma^2 + O_P(n^{-1/2}), \tag{125}$$

$$S_n^2 = \frac{n}{n-1}(M_n - \bar{X}_n^2) = \sigma^2 + O_P(n^{-1/2}). \tag{126}$$

The two first-order random remainders are absorbed into the same order; they are not canceled. The extra normalization contributes only a term of order the reciprocal sample size, because the bracket is tight. Finite second moment alone suffices for consistency by LLN; finite fourth moment is sufficient for the displayed rate. Simulation can illustrate the rate but does not replace checking these hypotheses.

Supplement: Connection to a one-step estimator (source p. 26, NOTE)

The variance expansion is already asymptotically linear:

$$S_n^2 - \sigma^2 = \frac{1}{n}\sum_i \{(X_i - \mu)^2 - \sigma^2\} + O_P(n^{-1}), \tag{127}$$

$$\sqrt{n}(S_n^2 - \sigma^2) \xrightarrow{d} N\left(0, \mathbb{E}(X_1 - \mu)^4 - \sigma^4\right). \tag{128}$$

This is the same first-order idea used in a one-step correction. For a smooth scalar estimating equation with nonzero population derivative, its Newton update from a preliminary estimate is

$$\Psi_n(t) = n^{-1}\sum_i \psi(X_i, t), \tag{129}$$

$$\hat{\theta}_{\text{one}} = \tilde{\theta} - \{\dot{\Psi}_n(\tilde{\theta})\}^{-1}\Psi_n(\tilde{\theta}). \tag{130}$$

With a consistent empirical derivative and controlled second derivatives, a preliminary error smaller than the inverse fourth root of sample size makes its quadratic Taylor remainder negligible on the root-sample-size scale. These are additional regularity requirements; order notation alone does not guarantee a valid one-step procedure. The variance identity above is exact up to an explicitly controlled remainder, without a Newton step.

9.5 Gaussian concentration gives more than a stochastic order

Student source: p. 27, "MGF + Markov's: how?" For IID centered normal observations with positive variance, both bounds below imply the usual sample-mean order:

$$\mathbb{P}(|\bar{X}_n| > \varepsilon) \le \frac{\sigma^2}{n\varepsilon^2}, \tag{131}$$

$$\mathbb{P}(|\bar{X}_n| > \varepsilon) \le 2\exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right), \tag{132}$$

$$\bar{X}_n = O_P(n^{-1/2}). \tag{133}$$

For the exponential bound, use the Gaussian moment generating function and Markov's inequality, for any positive tuning parameter:

$$\mathbb{E}e^{t\bar{X}_n} = \exp(t^2\sigma^2/(2n)), \tag{134}$$

$$\mathbb{P}(\bar{X}_n > \varepsilon) \leq \exp(-t\varepsilon + t^2\sigma^2/(2n)). \tag{135}$$

Minimize at the threshold times sample size divided by the variance and apply the same argument to the negative sample mean. The handwritten constant is therefore the reciprocal of twice the population variance. Substituting a cutoff times the inverse square root of sample size gives

$$\mathbb{P}(\sqrt{n}|\bar{X}_n| > M) \leq \min\{\sigma^2/M^2,\, 2e^{-M^2/(2\sigma^2)}\}. \tag{136}$$

Both bounds can be made arbitrarily small by increasing the cutoff, which proves stochastic boundedness. At a fixed positive threshold the exponential bound decays much faster in sample size. Stochastic order alone suppresses that quantitative distinction.

10 Delta method and asymptotic linearity

Missing from original typed notes: G5: all of Lecture 2 Section 2.6 was missing

Lecture 2, Theorems 2.29–2.31 and Examples 2.30–2.32, pp. 2-11–2-14. The updated student notes cover this material on pp. 28–31; it is not in the supplied teacher handnotes through October 21. The lecture's displayed conclusions omit the rate multiplier; its rate sequence must diverge, rather than decrease.

Student source: pp. 28–31; lecture Theorems 2.29–2.31.

Suppose a diverging deterministic rate yields a limiting distribution and the transformation is differentiable at the target:

$$r_n \to \infty, \qquad r_n(W_n - \omega_0) \xrightarrow{d} Z, \tag{137}$$

$$f(W_n) - f(\omega_0) = J_f(\omega_0)(W_n - \omega_0) + o_P(r_n^{-1}), \tag{138}$$

$$r_n\{f(W_n) - f(\omega_0)\} \xrightarrow{d} J_f(\omega_0)Z. \tag{139}$$

The proof writes the deterministic differentiability remainder as an error of order smaller than the input norm, uses consistency to make its coefficient small in probability, and tightness to bound the scaled input. For a normal limit the covariance transforms to

$$J_f(\omega_0)\Sigma J_f(\omega_0)^{\mathsf{T}}. \tag{140}$$

Supplement: Vector delta method and Jacobian dimensions (source p. 28)

For an input in a finite-dimensional Euclidean space and a vector-valued map, the Jacobian has one row per output and one column per input:

$$W_n, \omega_0 \in \mathbb{R}^d, \quad f : \mathbb{R}^d \to \mathbb{R}^k, \quad J_f(\omega_0) \in \mathbb{R}^{k\times d}, \tag{141}$$

$$r_n(W_n - \omega_0) \xrightarrow{d} N_d(0, \Sigma), \tag{142}$$

$$r_n\{f(W_n) - f(\omega_0)\} \xrightarrow{d} N_k\big(0, J_f(\omega_0)\Sigma J_f(\omega_0)^{\mathsf{T}}\big)\,. \tag{143}$$

For scalar output the gradient is a column, so its transpose is the Jacobian. The handwritten gradient-transpose covariance formula and the Jacobian formula above are the same convention expressed differently.

For exponential observations with rate parameter, the reciprocal sample mean has limit

$$\sqrt{n}(\bar{X}_n - \lambda^{-1}) \xrightarrow{d} N(0, \lambda^{-2}) \quad \Longrightarrow \quad \sqrt{n}(\bar{X}_n^{-1} - \lambda) \xrightarrow{d} N(0, \lambda^2). \tag{144}$$

If the first derivative vanishes, first-order delta gives a degenerate limit; a higher-order expansion may be needed. For example,

$$\sqrt{n}W_n \xrightarrow{d} N(0, \sigma^2) \quad \Longrightarrow \quad nW_n^2 \xrightarrow{d} \sigma^2\chi_1^2. \tag{145}$$

The last example is an editorial exercise, not a stated lecture theorem.

An asymptotically linear estimator has the representation

$$\hat{\theta}_n - \theta_0 = \frac{1}{n}\sum_i \psi(X_i) + o_P(n^{-1/2}), \qquad \mathbb{E}\psi(X_1) = 0, \quad \mathbb{E}\|\psi(X_1)\|^2 < \infty. \tag{146}$$

Its asymptotic covariance is the covariance of its influence function. For a transformed empirical mean the influence function is the Jacobian times the centered summand, not its uncentered second moment.

10.1 Relative risk: cell proportions, covariance, and inference

Student source: pp. 29–30; lecture Example 2.32. Assume IID binary observation pairs and equal treatment probabilities. The handwritten target is a ratio conditional on observing an outcome:

$$\mathbb{P}(T = 1) = \mathbb{P}(T = 0) = 1/2, \qquad (T, Y) \in \{0, 1\}^2, \tag{147}$$

$$\theta = \frac{\mathbb{P}(T = 1 \mid Y = 1)}{\mathbb{P}(T = 0 \mid Y = 1)} = \frac{\mathbb{P}(T = 1, Y = 1)}{\mathbb{P}(T = 0, Y = 1)}. \tag{148}$$

The outcome probability cancels in this expression. By Bayes' rule and equal allocation, it also equals the usual outcome risk ratio:

$$\theta = \frac{\mathbb{P}(Y = 1 \mid T = 1)}{\mathbb{P}(Y = 1 \mid T = 0)}. \tag{149}$$

Without equal allocation, the ratio of joint cells differs from that risk ratio by the allocation odds. Define the cell indicators and proportions:

$$\zeta_{ty} = \mathbb{P}(T = t, Y = y), \qquad a = \zeta_{11} > 0, \quad b = \zeta_{01} > 0, \tag{150}$$

$$A_i = T_i Y_i, \qquad B_i = (1 - T_i) Y_i, \tag{151}$$

$$A_n = n^{-1} \sum_i A_i, \qquad B_n = n^{-1} \sum_i B_i, \qquad \hat{\theta}_n = A_n / B_n, \quad \theta = a/b. \tag{152}$$

The positive denominator assumption ensures the estimator is defined with probability tending to one; set it arbitrarily when no denominator cell is observed. For the nondegenerate log and Wald formulas below we also assume a positive numerator cell probability.

TODO resolved: What are the entries of the covariance matrix? (source p. 29)

The cell indicators are Bernoulli, and their product is identically zero:

$$\operatorname{Var}(A_i) = a(1-a), \qquad \operatorname{Var}(B_i) = b(1-b), \tag{153}$$

$$A_i B_i = 0, \qquad \operatorname{Cov}(A_i, B_i) = -ab, \tag{154}$$

$$\Sigma = \begin{pmatrix} a(1-a) & -ab \\ -ab & b(1-b) \end{pmatrix}. \tag{155}$$

The off-diagonal entries are equal by symmetry of covariance. Disjoint cells are not independent: the negative covariance is essential.

Correction: Center the vector before applying the CLT (source p. 29)

The handwriting scales the uncentered vector. The correct statement is

$$\sqrt{n} \left\{ \begin{pmatrix} A_n \\ B_n \end{pmatrix} - \begin{pmatrix} a \\ b \end{pmatrix} \right\} \xrightarrow{d} N_2(0, \Sigma). \tag{156}$$

Without centering, the scaled mean vector diverges because its population cell probabilities are positive.

For the ratio map, the gradient and the delta-method variance are

$$f(x, y) = x/y, \qquad \nabla f(x, y) = \begin{pmatrix} 1/y \\ -x/y^2 \end{pmatrix}, \tag{157}$$

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, V), \tag{158}$$

$$V = \nabla f(a, b)^{\mathsf{T}} \Sigma \nabla f(a, b) = \frac{a}{b^2} + \frac{a^2}{b^3} = \frac{\theta(1+\theta)}{b}. \tag{159}$$

A consistent plug-in variance and an asymptotic Wald interval are

$$\widehat{V}_n = \frac{A_n}{B_n^2} + \frac{A_n^2}{B_n^3}, \tag{160}$$

$$\hat{\theta}_n \;\pm\; z_{1-\alpha/2} \sqrt{\widehat{V}_n / n}. \tag{161}$$

A log-scale interval stays positive after exponentiation:

$$\sqrt{n}\{\log \hat{\theta}_n - \log \theta\} \xrightarrow{d} N(0, 1/a + 1/b), \tag{162}$$

$$\text{interval} = \exp\left\{\log \hat{\theta}_n \pm z_{1-\alpha/2}\sqrt{(1/A_n + 1/B_n)/n}\right\}. \tag{163}$$

These approximations require positive observed cell proportions; they can be poor for sparse cells at a finite sample size.

10.2 Modern approach: expand the ratio directly

TODO resolved: Finish the teacher-note expansion (source p. 31)

The handwriting stops after substituting population values plus errors. Write the errors and apply the inverse expansion:

$$\Delta_{1n} = A_n - a = O_P(n^{-1/2}), \tag{164}$$
$$\Delta_{0n} = B_n - b = O_P(n^{-1/2}), \tag{165}$$
$$\hat{\theta}_n = \frac{a + \Delta_{1n}}{b + \Delta_{0n}}, \tag{166}$$
$$(b + \Delta_{0n})^{-1} = b^{-1} - b^{-2}\Delta_{0n} + O_P(n^{-1}), \tag{167}$$
$$\hat{\theta}_n - \theta = \frac{\Delta_{1n}}{b} - \frac{a\Delta_{0n}}{b^2} + O_P(n^{-1}). \tag{168}$$

The inverse remainder follows, on a neighborhood of the positive population denominator, from the exact identity

$$\frac{1}{b+d} = \frac{1}{b} - \frac{d}{b^2} + \frac{d^2}{b^2(b+d)}. \tag{169}$$

Products of two root-sample-size errors have reciprocal-sample-size order and disappear after multiplication by the square root of sample size. Applying the multivariate CLT to the remaining linear terms recovers the same variance as the delta method.

The corresponding influence function makes the cancellation transparent:

$$\psi(T_i, Y_i) = \frac{A_i - a}{b} - \frac{a(B_i - b)}{b^2} = \frac{A_i}{b} - \frac{aB_i}{b^2}, \tag{170}$$
$$\mathbb{E}\psi(T_i, Y_i) = 0, \qquad \mathrm{Var}\{\psi(T_i, Y_i)\} = V, \tag{171}$$
$$\hat{\theta}_n - \theta = n^{-1}\sum_i \psi(T_i, Y_i) + O_P(n^{-1}). \tag{172}$$

Here the population constants really cancel algebraically; generic stochastic-order symbols in the earlier variance calculation do not.

11 Bootstrap consistency with only a second moment

Missing from original typed notes: G6: the conditional triangular-array proof

Lecture 2, Appendix A.1, pp. 2-14–2-16. Original Notes 3 gives only the stronger third-moment Berry–Esseen argument.

Under a positive finite variance, conditionally on the observations,

$$\mathcal{L}^*\{\sqrt{n}(\bar{X}_n^* - \bar{X}_n)\} \Rightarrow N(0, \sigma^2) \quad \text{almost surely.} \tag{173}$$

Here is the key tail step behind conditional Lindeberg. For any fixed positive threshold and any fixed truncation level, eventually

$$\frac{1}{n}\sum_i X_i^2 \mathbf{1}\{|X_i| > \varepsilon\sqrt{n}\} \le \frac{1}{n}\sum_i X_i^2 \mathbf{1}\{|X_i| > K\}, \tag{174}$$

$$\limsup_n \frac{1}{n}\sum_i X_i^2 \mathbf{1}\{|X_i| > \varepsilon\sqrt{n}\} \le \mathbb{E}[X_1^2 \mathbf{1}\{|X_1| > K\}] \longrightarrow 0 \quad (K \to \infty). \tag{175}$$

Centering by the sample mean does not alter this conclusion because the sample mean converges to a finite constant. Conditional total variance converges to the population variance by LLN. Apply the triangular-array CLT on probability-one sample paths, then Pólya's theorem for the continuous normal limit to obtain conditional KS consistency. This proves convergence without the third-moment rate. It does not assert an automatic rate for every bootstrap confidence interval.

A Illustrated distributional distances

Editorial supplement based on lecture Appendix B, pp. 2-17–2-23.

Each distance asks a different question about two probability laws. For real-valued variables write their CDFs as follows:

$$F(t) = P((-\infty, t]), \qquad G(t) = Q((-\infty, t]). \tag{176}$$

Lévy and KS use the CDF directly; Prokhorov uses neighborhoods of events; BL and Wasserstein use test functions or transport; TV and Hellinger use probability mass or density geometry. Every figure below illustrates a definition or an exactly computable example.

A.1 Lévy: allow horizontal and vertical CDF error

$$d_L(F, G) = \inf\{\varepsilon > 0 : F(t - \varepsilon) - \varepsilon \le G(t) \le F(t + \varepsilon) + \varepsilon \text{ for all } t\}. \tag{177}$$

Intuition. Enclose one CDF between shifted copies of the other. Horizontal slack allows nearby locations to count as similar; vertical slack tolerates a small probability discrepancy. Both use the same numerical budget, so the value depends on the units of the horizontal axis.

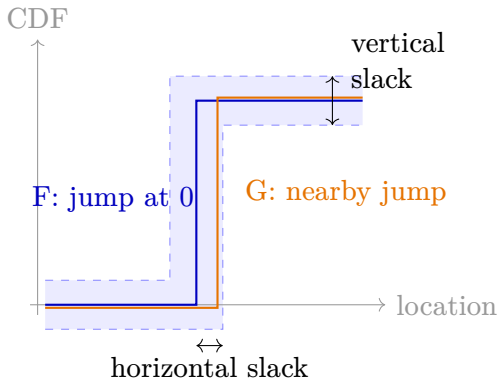


Figure A.1. Blue dashed curves form a horizontal-and-vertical slack envelope around a unit step. The orange step lies inside the envelope. Horizontal and vertical scales are intentionally different for legibility.

Exact example. Moving a unit point mass by a nonnegative displacement gives

$$d_L(\delta_0, \delta_h) = \min\{h, 1\}, \qquad h \ge 0. \tag{178}$$

A small shift is cheap even though the CDF jumps. This is why the metric can go to zero at a discontinuous limiting law. For a fixed limiting law on the real line, vanishing Lévy distance is equivalent to weak convergence.

A.2 Prokhorov: enlarge an event and allow missing mass

$$A^\varepsilon = \{x : d(x, A) < \varepsilon\}, \tag{179}$$

$$d_{\mathrm{Pr}}(P, Q) = \inf\{\varepsilon > 0 : \quad P(A) \le Q(A^\varepsilon) + \varepsilon, \quad Q(A) \le P(A^\varepsilon) + \varepsilon \text{ for every Borel } A\}. \tag{180}$$

Intuition. Before comparing an event's probability, expand its boundary by the spatial budget. Then allow an additional probability budget of unmatched mass. This accommodates both small spatial errors and rare large errors.

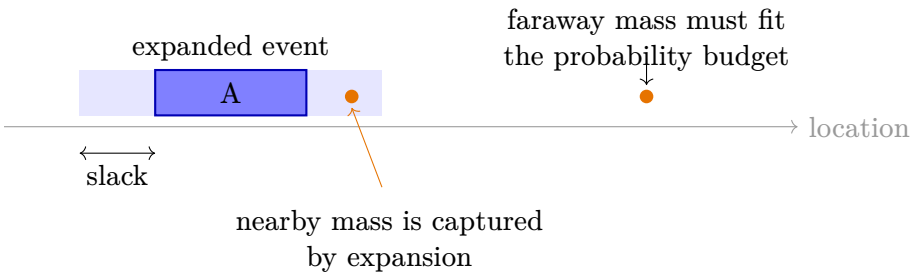


Figure A.2. A set and its open neighborhood. The nearby orange point is recovered geometrically; the distant point must be accounted for as unmatched probability.

Coupling interpretation. On a Euclidean space, this metric is the infimum of budgets for which a coupling of the two laws can satisfy

$$\mathbb{P}(\|X - Y\| > \varepsilon) \le \varepsilon, \qquad X \sim P, \quad Y \sim Q. \tag{181}$$

Strict versus non-strict inequalities at the boundary do not change the infimum. This does not require the originally observed variables to have this joint distribution; one may choose any coupling.

For point masses and a contamination example,

$$d_{\mathrm{Pr}}(\delta_0, \delta_h) = \min\{|h|, 1\}, \tag{182}$$

$$Q = (1 - a)\delta_0 + a\delta_M \quad \Longrightarrow \quad d_{\mathrm{Pr}}(\delta_0, Q) \le a, \qquad 0 \le a \le 1. \tag{183}$$

A rare outlier is cheap regardless of how far away it lies. The metric metrizes weak convergence in Euclidean spaces; it does not impose moment convergence.

A.3 Bounded Lipschitz: limited range and slope

$$d_{\mathrm{BL}}(P,Q) = \sup_{\|f\|_\infty \le 1,\, \mathrm{Lip}(f)\le 1} |\mathbb{E}_P f - \mathbb{E}_Q f|. \tag{184}$$

Intuition. Let an observer choose a scoring function. The function may change by at most the distance between inputs and must stay between minus one and one. The best observer's expected-score gap is the distance. Both constraints matter: slope limits sensitivity to nearby shifts; range limits sensitivity to distant outliers.

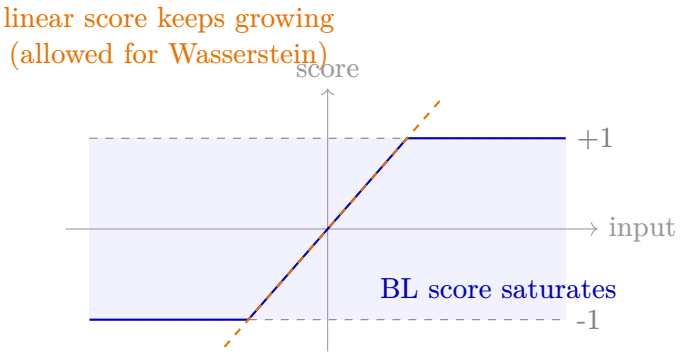


Figure A.3. The clipped identity is a valid BL score. The unbounded identity satisfies the slope constraint but fails the BL range constraint.

For two point masses the optimal score can use the full difference between minus one and one:

$$d_{\mathrm{BL}}(\delta_x, \delta_y) = \min\{\|x-y\|, 2\}. \tag{185}$$

For a small mixture perturbation,

$$Q = (1-a)P + aR \quad \Longrightarrow \quad d_{\mathrm{BL}}(P,Q) \le 2a. \tag{186}$$

The factor two comes from the possible difference of two scores in the allowed range. In particular, a bad event with vanishing probability cannot prevent BL convergence even if its magnitude explodes.

Definition convention

We follow the lecture's separate range and slope bounds. Some books instead bound the sum of these two quantities by one. These conventions induce the same weak convergence, but numerical distances and comparison constants can differ. A supremum over the full fixed class is stronger than checking one chosen score; both characterize weak convergence when used correctly.

A.4 Kolmogorov–Smirnov: the worst threshold-probability error

$$d_{\mathrm{KS}}(F, G) = \sup_t |F(t) - G(t)| = \sup_t |P(X \le t) - Q(Y \le t)|. \tag{187}$$

Intuition. Place a threshold anywhere on the number line. Compare the probability to its left under the two laws. KS is the largest vertical CDF gap; there is no permission to shift the threshold horizontally.

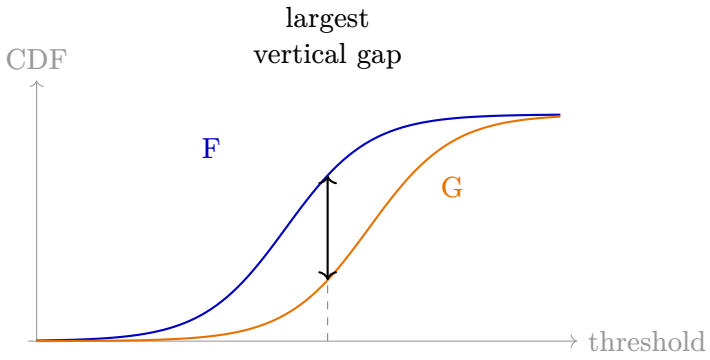


Figure A.4. Two logistic CDFs shifted by one unit. The maximum vertical separation occurs halfway between their centers.

KS bounds directly control probability error for a one-sided event. For an interval, subtract two CDFs, giving at most twice the KS error. This explains its role in Berry–Esseen bounds and confidence interval coverage.

An arbitrarily small shift of a point mass gives

$$d_{\mathrm{KS}}(\delta_0, \delta_h) = 1 \quad (h \neq 0), \qquad d_L(\delta_0, \delta_h) \to 0 \quad (h \to 0). \tag{188}$$

A threshold placed between the two atoms separates them perfectly. Hence KS is strictly stronger than weak convergence in general. At a fixed continuous limiting CDF, Pólya's theorem makes the two convergence criteria equivalent. This is an asymptotic statement, not a universal numerical bound between the two distances.

A.5 Wasserstein: minimum transport cost

For a real exponent at least one and laws with finite corresponding moments,

$$W_p(P,Q) = \inf_{\pi \in \Pi(P,Q)} \left(\int \|x-y\|^p \, d\pi(x,y) \right)^{1/p}. \tag{189}$$

Intuition. Treat probability as divisible material. A transport plan specifies how much mass goes from each origin to each destination. Choose the plan with least average powered distance, then take the root. The joint law is a coupling; it may split mass and need not be a deterministic transport map. Larger powers penalize long moves more heavily.

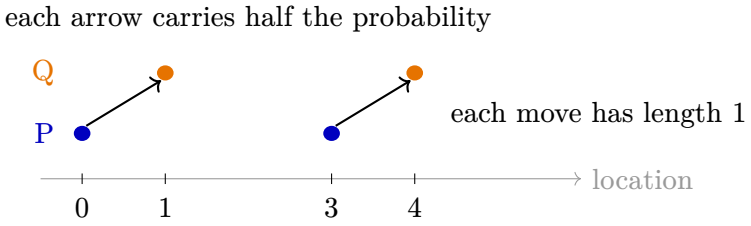


Figure A.5. Equal masses at zero and three move to one and four. The monotone matching is optimal in one dimension, and every Wasserstein distance with exponent at least one equals one in this example.

In one dimension the common-quantile coupling is optimal:

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}, \qquad 0 < u < 1, \tag{190}$$

$$W_p^p(P,Q) = \int_0^1 |F^{-1}(u) - G^{-1}(u)|^p \, du, \tag{191}$$

$$W_1(P,Q) = \int_{\mathbb{R}} |F(t) - G(t)| \, dt = \sup_{\mathrm{Lip}(f) \leq 1} |\mathbb{E}_P f - \mathbb{E}_Q f|. \tag{192}$$

Thus KS measures the tallest CDF gap, whereas one-dimensional Wasserstein with exponent one measures the *area* between CDFs. For a rare outlier,

$$Q = (1-a)\delta_0 + a\delta_M \quad \Longrightarrow \quad W_p(\delta_0, Q) = a^{1/p}|M|. \tag{193}$$

Unlike BL or Prokhorov, Wasserstein charges for how far the outlier travels. Its convergence criterion in Euclidean space is

$$W_p(P_n, P) \to 0 \quad \Longleftrightarrow \quad P_n \Rightarrow P \text{ and } \int \|x\|^p \, dP_n \to \int \|x\|^p \, dP < \infty. \tag{194}$$

Distinguish two notions. Moment convergence of given random variables uses their actual joint law; Wasserstein minimizes over all joint laws. Thus moment convergence implies Wasserstein convergence of the laws, but the converse need not hold for the original coupling. Uniform integrability is another route from weak convergence to moment convergence; using a Wasserstein metric is not mandatory.

A.6 Total variation: the best event for distinguishing two laws

For densities with respect to a common dominating measure,

$$p = \frac{dP}{d\lambda}, \qquad q = \frac{dQ}{d\lambda}, \tag{195}$$

$$d_{\mathrm{TV}}(P, Q) = \sup_{A \text{ Borel}} |P(A) - Q(A)| \tag{196}$$

$$= \frac{1}{2} \int |p - q| \, d\lambda = 1 - \int \min(p, q) \, d\lambda. \tag{197}$$

Intuition. Choose *any* measurable event to maximize the probability gap. A maximizing event is the region where the first density exceeds the second. The distance is the probability mass left over after overlapping as much of the two laws as possible. It ignores spatial distance.

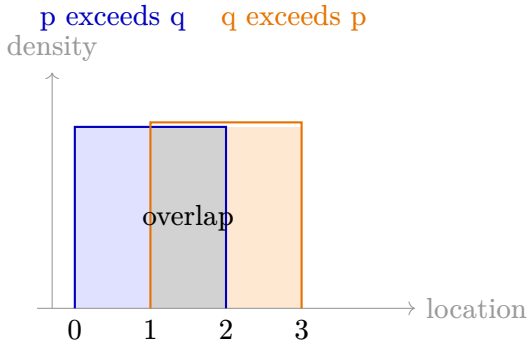


Figure A.6. Uniform laws on the intervals from zero to two and from one to three. Both density heights are one half; the drawing rescales the vertical axis. The overlap has mass one half, so TV equals one half. The orange top edge is offset slightly to show both curves.

Testing interpretation. If either law is equally likely a priori, the minimum average probability of classification error is

$$R^*_{\mathrm{test}} = \frac{1 - d_{\mathrm{TV}}(P, Q)}{2}. \tag{198}$$

Zero TV means the laws cannot be distinguished; TV equal to one means there is a measurable event that perfectly distinguishes them. Under an optimal coupling, TV is also the minimum probability that the two coupled variables are unequal.

A point mass moved by any nonzero amount has TV equal to one, even if the move is tiny. By contrast, a contamination of size one percent changes any event probability by at most one percent:

$$d_{\mathrm{TV}}(P, (1 - a)P + aR) \leq a. \tag{199}$$

A density with respect to Lebesgue measure is unnecessary: taking the dominating measure to be the sum of the two probability laws always works. This covers discrete, continuous, and mixed distributions uniformly.

A.7 Hellinger: square-root density geometry

We use the normalized convention from lecture Definition 2.51:

$$H^2(P,Q) = \frac{1}{2}\int (\sqrt{p}-\sqrt{q})^2\, d\lambda = 1-\rho(P,Q), \tag{200}$$

$$\rho(P,Q) = \int \sqrt{pq}\, d\lambda, \qquad 0 \le H(P,Q) \le 1. \tag{201}$$

Intuition. Taking the square root of a density maps every law to a unit vector in an inner-product space. Affinity is the inner product of the two unit vectors. Hellinger distance is their Euclidean chord length divided by the square root of two, not the angular or arc distance.

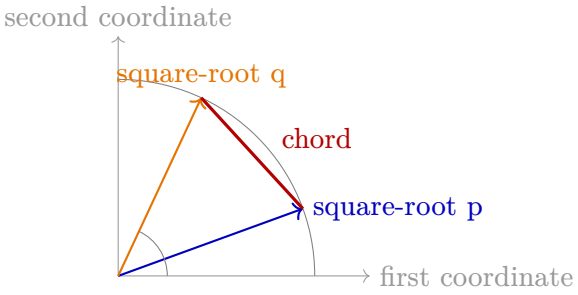


Figure A.7. For two-point distributions, square-root probability vectors lie on the positive quadrant of a unit circle. General densities have the same geometry in an infinite-dimensional function space.

For Bernoulli laws, the formula is explicit:

$$H^2(\mathrm{Ber}(a),\mathrm{Ber}(b)) = 1-\sqrt{ab}-\sqrt{(1-a)(1-b)}. \tag{202}$$

Affinity multiplies across independent observations, so

$$H^2(P^{\otimes n},Q^{\otimes n}) = 1-\rho(P,Q)^n = 1-(1-H^2(P,Q))^n. \tag{203}$$

This makes Hellinger useful for likelihood theory and testing lower bounds. In a regular differentiable parametric family, a small parameter displacement has the local Fisher-information approximation

$$H^2(P_{\theta+h},P_\theta) = \frac{1}{8}h^{\mathsf{T}} I(\theta) h + o(\|h\|^2). \tag{204}$$

This approximation requires quadratic-mean differentiability; it is not an identity valid for every family.

Correction: Hellinger normalization in the lecture

Definition 2.51 uses the normalized convention above. The upper TV bound in Theorem 2.52 and the product formula in Appendix B.4 mix conventions. With this document's convention the consistent formulas are

$$H^2 \le d_{\mathrm{TV}} \le H\sqrt{2-H^2}, \qquad H^2(P^{\otimes n},Q^{\otimes n}) = 1-(1-H^2(P,Q))^n. \tag{205}$$

The alternative unnormalized distance equals the normalized distance times the square root of two. All constants must change together.

A.8 What converges, and what may still go wrong?

The following comparison uses a fixed target law. TV and Hellinger have equivalent notions of convergence; all moments mentioned are finite.

Distance	Question it asks	What vanishing distance guarantees
Lévy	How much CDF shift and vertical slack?	Weak convergence on the real line.
Prokhorov	How much event expansion and unmatched mass?	Weak convergence in Euclidean spaces.
BL	Largest gap for bounded smooth scores?	Weak convergence in Euclidean spaces.
KS	Largest threshold probability error?	Weak convergence; uniform CDF control.
Wasserstein	Smallest powered transport cost?	Weak convergence and convergence of the corresponding absolute moment.
TV	Largest discrepancy over all events?	Uniform control of every measurable event.
Hellinger	Separation of square-root densities?	TV convergence, with useful product geometry.

For the lecture's BL convention, useful correct inequalities include

$$\tfrac{1}{3} d_{\mathrm{BL}} \le d_{\mathrm{Pr}} \le \sqrt{d_{\mathrm{BL}}}, \tag{206}$$

$$d_L \le d_{\mathrm{KS}} \le d_{\mathrm{TV}}, \qquad d_L \le d_{\mathrm{Pr}} \le d_{\mathrm{TV}}, \tag{207}$$

$$d_{\mathrm{BL}} \le W_1 \le W_p \quad (p \ge 1), \tag{208}$$

$$d_{\mathrm{Pr}} \le W_p^{p/(p+1)}, \qquad H^2 \le d_{\mathrm{TV}} \le H\sqrt{2 - H^2}. \tag{209}$$

The Lévy comparisons apply in one dimension. The Wasserstein inequalities use the same ground metric and finite moments. The Prokhorov bound follows by choosing an almost optimal transport coupling and applying Markov's inequality to its distance.

Correction: BL–Prokhorov constants depend on the convention

For the separate range and slope bounds in Definition 2.38, the lower bound in lecture Theorem 2.52 must be weakened as above. To check the constant directly, under a coupling with distance exceeding a budget on at most that budget of probability, the expected score difference is at most three times the budget: one contribution from nearby points and two from the exceptional event.

The upper bound follows by testing a neighborhood with the ramp function

$$f(x) = \max\{1 - d(x, A)/r, 0\}, \qquad 0 < r \le 1. \tag{210}$$

The function scaled by the radius meets both BL constraints, giving

$$P(A) - Q(A^r) \le d_{\mathrm{BL}}(P, Q)/r. \tag{211}$$

Choose the radius equal to the square root of BL distance when that distance is at most one. For larger values, Prokhorov is at most one.

A.9 Two examples that separate the distances

Example 1: a nearby atom. For a positive displacement smaller than one,

$$P = \delta_0, \quad Q = \delta_h : \qquad d_L = d_{\mathrm{Pr}} = d_{\mathrm{BL}} = W_p = h, \tag{212}$$

$$d_{\mathrm{KS}} = d_{\mathrm{TV}} = H = 1. \tag{213}$$

Metrics that tolerate spatial displacement regard the laws as close. Metrics that can test exact support regard them as completely separated.

Example 2: rare, faraway contamination. Take

$$P_n = (1 - 1/n)\delta_0 + (1/n)\delta_{n^2}, \qquad P = \delta_0. \tag{214}$$

Then the small contamination mass implies

$$d_{\mathrm{KS}}(P_n, P) = d_{\mathrm{TV}}(P_n, P) = 1/n \to 0, \tag{215}$$

$$H^2(P_n, P) = 1 - \sqrt{1 - 1/n} \to 0, \tag{216}$$

$$d_{\mathrm{BL}}(P_n, P) \le 2/n \to 0, \qquad d_{\mathrm{Pr}}(P_n, P) \le 1/n \to 0. \tag{217}$$

Nevertheless, every unit of contaminated mass must travel back to zero:

$$W_p(P_n, P) = n^{2-1/p} \longrightarrow \infty \qquad (p \ge 1). \tag{218}$$

Thus even TV convergence does not guarantee moment convergence or Wasserstein convergence on an unbounded space. The reverse implication also fails: nearby point masses have vanishing Wasserstein distance and TV identically one. KS convergence and Wasserstein convergence are likewise incomparable without additional assumptions.

A.10 A fixed target matters: topology versus uniform equivalence

Lecture Example 2.53.

Consider equal weights on consecutive integers and on their half-unit shifts:

$$P_n = \frac{1}{n}\sum_{j=1}^{n} \delta_j, \qquad Q_n = \frac{1}{n}\sum_{j=1}^{n} \delta_{j+1/2}. \tag{219}$$

Their CDFs differ by at most the height of a single jump:

$$d_L(P_n, Q_n) \le d_{\mathrm{KS}}(P_n, Q_n) = 1/n \to 0, \qquad d_{\mathrm{Pr}}(P_n, Q_n) = 1/2. \tag{220}$$

For the lower bound on Prokhorov, choose the set of integer support points. A neighborhood with radius at most one half contains no shifted support point, so a smaller probability budget cannot cover its unit mass. The natural half-unit matching proves the upper bound by taking radii just above one half.

There is no contradiction: these are two moving, nontight sequences, not convergence to a fixed target law. Lévy and Prokhorov induce the same weak topology but are not uniformly equivalent on the entire space of real probability laws. BL and Prokhorov have uniform comparison bounds.